Jie Liu

Human-AI Interaction | Computer Vision | Physical AI | Embodied Agents

Amsterdam / Eindhoven, Netherlands | j.liu9@tue.nl | jliu4ai.github.io | GitHub: jliu4ai | LinkedIn: jie-liu-390664250

Research Profile

I am a Postdoc Researcher at **Eindhoven University of Technology (TU/e), AMOR/e Lab**, working with Prof. **Joaquin Vanschoren** on LLMs and MLLMs through the OpenEuroLLM project. I obtained my PhD from **VISLab, University of Amsterdam**, supervised by Prof. **Efstratios Gavves**. My research builds human-centered AI for perception, reasoning, and interaction in the physical world, spanning computer vision, multimodal reasoning, 3D human-scene interaction, interactive segmentation, and embodied agents. I also serve as a regular reviewer for CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, T-PAMI, and IJCV.

Research Experience

Eindhoven University of Technology (TU/e), AMOR/e Lab — Nov. 2025 – Present
Postdoc Researcher — *Eindhoven, Netherlands*

- Contributing to the **OpenEuroLLM** project, focusing on LLM and MLLM post-training, including parameter-efficient fine-tuning, instruction tuning, and reinforcement learning.
- Conducting research on multimodal reasoning, physical reasoning, and embodied agents, building AI systems that perform perception, reasoning, and interaction in the physical world.

Meshcapade, spin-off from Max Planck Institute for Intelligent Systems — May 2025 – Nov. 2025
ML Research Scientist Intern — *Tubingen / Remote*

- Worked with Dr. **Yan Zhang** and Prof. **Michael J. Black** on SMPL-X-based digital humans.
- Built FunHSI, an automated system for open-vocabulary 3D human-scene interaction generation from prompts and posed RGB-D images, enabling functional interactions in 3D environments.

University of Amsterdam, VISLAB / ELLIS — Sep. 2021 – Oct. 2025
Ph.D. Researcher in Computer Science — *Amsterdam, Netherlands*

- Conducted research on human-centered visual segmentation, including robust perception under sparse supervision, user interaction, and domain shifts, with applications to 2D images, 3D point clouds, and vision-language models.

Highlighted Research

Selective Latent Thinking: Adaptive Compression of LLM Reasoning Chains LLM Reasoning

Long chain-of-thought reasoning improves complex problem solving but often introduces substantial computational overhead through redundant intermediate steps. This project investigates adaptive reasoning-chain compression, representing low-information spans in latent space while preserving explicit tokens for precision-critical reasoning, with the goal of improving the accuracy-efficiency trade-off in LLM inference.

Open-Vocabulary Functional 3D Human-Scene Interaction Generation Human-Scene Interaction

Functional 3D human-scene interaction generation requires joint reasoning over language intent, object affordances, scene geometry, and physically plausible human motion. FunHSI is a training-free framework that integrates open-vocabulary prompts, posed RGB-D observations, SMPL-X body representations, and motion generation to synthesize semantically meaningful and functionally valid human-scene interactions.

Probabilistic Interactive 3D Segmentation with Hierarchical Neural Processes 3D Computer Vision

Interactive 3D segmentation remains challenging because models must infer object boundaries from sparse user prompts while generalizing across diverse point-cloud geometries. This project addresses this problem with a hierarchical neural-process framework that models segmentation as probabilistic function prediction, enabling uncertainty-aware interaction and few-shot generalization to unseen objects.

Cooperative Plan Optimization for Efficient Embodied Multi-Agent Cooperation Embodied Agents

Multi-agent embodied cooperation requires high-level plans that account for shared goals, environmental constraints, and inter-agent dependencies. CaPo introduces an LLM-based cooperative plan optimization framework that refines agent plans through coordination-aware reasoning, improving collaboration efficiency between embodied agents.

Research Areas & Technical Skills

Core Research: LLM/MLLM reasoning; multimodal learning; physicla reasoning; vision-language-action models;.
Modeling Methods: post-training; reinforcement learning; diffusion and autoregressive models; hierarchical neural processes; prototype learning; optimal transport; graph-based reasoning.
Technical Stack: Python; PyTorch; Linux; slurm; Git; LaTeX; SMPL-X; RGB-D and point-cloud processing.

Education

University of Amsterdam, Amsterdam, Netherlands — Sep. 2021 – Sep. 2025
Ph.D. in Computer Science; ELLIS Ph.D. program, VISLAB

Northeastern University, Shenyang, China — Sep. 2018 – Jun. 2021
M.S. in Electrical Engineering

Selected Publications

1. **Selective Latent Thinking: Adaptive Compression of LLM Reasoning Chains** — arXiv 2026
   Hui Xie*, ***Jie Liu****, Ziyue Qiao, Joaquin Vanschoren. (*Equal contribution)
2. **Open-Vocabulary Functional 3D Human-Scene Interaction Generation** — arXiv 2026
   Jie Liu, Yu Sun, Alpar Cseke, Yao Feng, Nicolas Hron, Michael J. Black, Yan Zhang.
3. **Towards Uniformity and Alignment for Multimodal Representation Learning** — ICML 2026
   Wenzhe Yin, Pan Zhou, Zehao Xiao, ***Jie Liu***, Shujian Yu, Jan-Jakob Sonke, Efstratios Gavves.
4. **Probabilistic Prototype Calibration of VLMs for Generalized Few-shot Segmentation** — ICCV 2025
   Jie Liu, Jiayi Shen, Pan Zhou, Jan-Jakob Sonke, Efstratios Gavves.
5. **Probabilistic Interactive 3D Segmentation with Hierarchical Neural Processes** — ICML 2025
   Jie Liu, Pan Zhou, Zehao Xiao, Jiayi Shen, Wenzhe Yin, Jan-Jakob Sonke, Efstratios Gavves.
6. **Cooperative Plan Optimization for Efficient Embodied Multi-agent Cooperation** — ICLR 2025
   Jie Liu, Pan Zhou, Yingjun Du, Ah-Hwee Tan, Cees G. M. Snoek, Jan-Jakob Sonke, Efstratios Gavves.
7. **Click Prompt Learning with Optimal Transport for Interactive Segmentation** — ECCV 2024
   Jie Liu, Haochen Wang, Wenzhe Yin, Jan-Jakob Sonke, Efstratios Gavves.
8. **Dynamic Prototype Adaptation with Distillation for Few-shot Point Cloud Segmentation** — 3DV 2024
   Jie Liu, Wenzhe Yin, Haochen Wang, Yunlu Chen, Jan-Jakob Sonke, Efstratios Gavves.
9. **Domain Adaptation with Cauchy-Schwarz Divergence** — UAI 2024
   Wenzhe Yin, Shujian Yu, Yicong Lin, ***Jie Liu***, Jan-Jakob Sonke, Efstratios Gavves.
10. **PiClick: Picking the Desired Mask in Click-based Interactive Segmentation** — TMM 2024
    Cilin Yan, Haochen Wang, ***Jie Liu***, Xiaolong Jiang, Yao Hu, Xu Tang, Guoliang Kang, Efstratios Gavves.
11. **Few-shot Semantic Segmentation with Support-Induced Graph Convolutional Network** — BMVC 2022
    Jie Liu, Yanqi Bao, Haochen Wang, Wenzhe Yin, Jan-Jakob Sonke, Efstratios Gavves.
12. **Dynamic Prototype Convolution Network for Few-shot Semantic Segmentation** — CVPR 2022
    Jie Liu, Yanqi Bao, Guosen Xie, Huan Xiong, Jan-Jakob Sonke, Efstratios Gavves.
13. **Dynamic Transformer for Few-shot Instance Segmentation** — ACM MM 2022
    Haochen Wang, ***Jie Liu***, Yongtuo Liu, Subhransu Maji, Jan-Jakob Sonke, Efstratios Gavves.
14. **Scale-aware Graph Neural Network for Few-shot Semantic Segmentation** — CVPR 2021
    Guo-sen Xie*, ***Jie Liu****, Huan Xiong, Ling Shao. (*Equal contribution)